BUSINESS REQUIREMENTS DOCUMENT

Eratex End-to-End Planning & Scheduling Tool

Aligned with Rajesh Inputs and BRD Diff Recommendations

Field	Value
Company	Eratex
Document Type	Business Requirements Document
Product Scope	Denim-aware planning and scheduling platform for garment manufacturing operations
Manufacturing Scope	Denim bottoms and chinos; denim treated as dominant complexity driver
Baseline	Eratex_Planning_Scheduling_Tool_BRD_Professional_Generated.docx
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Status	Draft for business review

Core revision: The BRD now frames the product as a governed, denim-aware operating spine rather than only a scheduling screen or Excel replacement.

1. Executive Summary

Eratex operates as an export-oriented garment manufacturer serving international customers, primarily across denim bottoms and chinos. The production environment is not a simple cut-and-sew flow. For denim, commercial value is often created through dry processing, wet washing, fading, distressing, tearing, touch-up, repeat wash cycles, and buyer-specific shade/effect outcomes.

Although a planning tool such as FastReact may exist in the operating environment, planning still appears to depend heavily on Excel and manual coordination. The deeper issue is not only tool adoption. The deeper issue is that the real operating complexity of denim production - especially wash/laundry, rework, WIP movement, capacity changes, order changes, and shipment-risk recovery - is not governed by one trusted, live operating spine.

The proposed solution should therefore be an end-to-end, denim-aware planning and scheduling platform that converts customer demand into executable production control. It must support order visibility, PCD readiness, capacity-aware planning, daily release, sewing line loading, wash and rewash planning, WIP ageing, exceptions, recovery actions, and shipment readiness.

Business outcome: Maintain high OTIF, improve utilization, reduce overtime and firefighting, reduce Excel dependency, and protect shipment commitments through early, constraint-aware control.

2. Business Context

2.1 Export-Oriented Operating Context

Eratex should be treated as an export-oriented garmenting operation where customer delivery commitments, buyer approvals, inspection readiness, documentation, packing, and dispatch reliability are central to business performance. Orders are expected to be driven by international customers, with US and Europe referenced as primary markets in the consultant input.

2.2 End-to-End Garment Flow

- Customer enquiry, quotation visibility, sampling, buyer approval, and order confirmation.
- Fabric and trim procurement, fabric inward, fabric QC, shade-lot control, and PCD readiness.
- Cutting, sewing, dry process, wet wash, repeat wash or rework, finishing, final QC, packing, documentation, and shipment.

2.3 Denim-Specific Manufacturing Complexity

Denim manufacturing has a different complexity profile from basic garmenting. Visible effects such as fading, distressing, tearing, whiskering, scraping, grinding, uneven shade, hand feel, and fashion-wash outcomes may be intentional value-add features rather than defects.

This makes wash/laundry a value-creation, quality-risk, and capacity-constraining production domain. Dry process, wet wash, repeat wash cycles, post-wash shade/effect checks, and rewash loops must be planned explicitly because they affect capacity, WIP ageing, recovery cost, and shipment risk.

2.4 Chinos as Lower-Complexity Variant

Chinos should be supported through the same operating spine, but with simpler route and wash complexity unless business evidence proves otherwise.

3. Reconciled Problem Statement

Eratex is a large export-oriented denim and chinos garment manufacturer where production complexity extends beyond cutting and sewing into customer approvals, nominated material procurement, fabric QC, PCD readiness, sewing line loading, dry/wet wash processing, repeat wash cycles, quality holds, WIP movement, finishing, packing, and shipment readiness.

Although a formal planning tool may exist, the factory appears to rely heavily on Excel and manual coordination because the real operating complexity - especially denim wash/rework, shifting capacity constraints, order changes, and exception recovery - is not fully governed by one trusted system.

The factory is able to maintain OTIF above 95%, but this may be achieved through extra operating expense, overtime, expediting, resequencing, and manual firefighting, while resource utilization remains around 70%.

Core problem: The core problem is not the absence of software. It is the absence of a live, capacity-aware, wash-aware, WIP-trusted, exception-owned operating spine that converts customer demand into executable production control and reacts when factory reality changes.

3.1 Symptoms Observed in Current Planning

- Planning may be formally digital but operationally manual.
- OTIF is maintained through firefighting and cost absorption.
- Utilization is low because flow is unstable.
- Workcenter constraints shift without timely visibility.
- Wash rework and repeat cycles may not be fully capacity-planned.
- PCD readiness may not be enforced as a hard gate.
- WIP ageing may remain hidden between departments.
- Shipment readiness may be assessed too late.

3.2 Root Cause Behind the Symptoms

Excel remains active because the formal planning environment may not yet represent the real production states, wash/rework behaviour, capacity-change reactions, and exception ownership required by daily operations.

4. Business Need and Operating Spine

Eratex requires an integrated planning and scheduling tool that becomes the single operating layer for production control. It must answer daily readiness, bottleneck, WIP, exception, ownership, and shipment-risk questions.

Business Question	Required System Response
Which orders are ready or blocked?	Show lifecycle, PCD, material, wash, quality, and shipment readiness status.
Where is the bottleneck?	Show load, queue, WIP ageing, utilization, and constraint flags by workcenter.
What can be released today?	Validate material, prior-process, capacity, quality, manpower, and next-process readiness.
What recovery action is required?	Recommend rule-based actions with owner, due date, and approval requirement.
What changed in the plan?	Maintain version, audit trail, boundary-case event, and impact preview.

The product must also define how the system reacts when operating reality changes:

- What happens if an order is cancelled?
- What happens if capacity is lost or added?
- What happens if wash must repeat?
- What happens if quality fails after wash?
- What happens if shipment date is pulled in?
- What happens if a frozen weekly plan needs to change?
- What is recommended, what is blocked, and what requires approval?

5. Business Objectives and Success Metrics

5.1 Primary Objectives

1. Establish one integrated planning view from order confirmation to shipment.
2. Enforce PCD readiness before cutting starts.
3. Convert weekly production plans into daily executable releases.
4. Monitor total workcenter load and identify shifting constraints.
5. Improve sewing line loading using realistic capacity assumptions.
6. Treat denim wash/laundry as a value-creation, quality-risk, and capacity-constraining production domain, while supporting chinos as a simpler configured route variant where applicable.
7. Make WIP ageing visible across departments.
8. Track exceptions with owner, severity, due date, and recovery action.
9. Distinguish production completion from true shipment readiness.
10. Reduce reliance on Excel-based planning and manual firefighting.

11. Define scheduling boundary-case behaviour for cancellation, capacity change, rewash, quality failure, shipment pull-in, and plan change.
12. Establish the MVP planning grain so business users, planners, and builders share the same level of scheduling detail.
13. Produce a Scheduling Behaviour Rulebook before deep planning-engine implementation.

5.2 Success Metrics

Metric	Target Direction
OTIF	Maintain above 95% while reducing recovery cost.
Utilization	Improve from around 70% through better flow and readiness.
Planning adherence	Improve weekly and daily plan compliance.
PCD readiness accuracy	Reduce cutting releases with open blockers.
WIP ageing	Reduce ageing at key handover points.
Wash queue ageing	Reduce sewn garments waiting for wash.
Rewash visibility	Capture 100% of rewash cycles with reason, quantity, and capacity impact.
Boundary-case response	Cancellation, capacity change, and shipment pull-in events have visible owner and action.
Capacity definition completeness	100% of MVP workcenters have agreed capacity unit, bucket, and recovery levers.
Rulebook completion	Scheduling Behaviour Rulebook approved before full planning-engine build.

6. Product Scope and MVP Boundaries

The ten MVP surfaces remain valid. MVP scope should also capture upstream milestone visibility for customer enquiry, quotation status, sample submission, sample approval, technical file receipt, and order confirmation. Full quotation costing and commercial feasibility workflow should remain outside MVP unless confirmed by the business team.

6.1 Product Complexity Classes

Complexity Class	Planning Implication
DENIM_HEAVY_WASH	Dry/wet route, fashion effects, higher rewash risk, shade/effect QC mandatory.
DENIM_BASIC_WASH	Standard wash route, lower rewash probability.
CHINO_BASIC_WASH	Simpler route, lower wash complexity.
CHINO_NO_COMPLEX_WASH	Standard garmenting route without denim-style wash complexity.

6.2 MVP Surfaces

#	Surface	Primary Role in MVP
1	Order Lifecycle	One digital thread and upstream milestone visibility.
2	PCD Readiness	Gate-based readiness before cutting.
3	Weekly Planning Workbench	Capacity-checked coordination contract.
4	Daily Production Release	Executable release based on true readiness.
5	Workcenter Load Monitor	Load, queue, capacity, WIP, and bottleneck visibility.
6	Sewing Line Loading	Realistic line targets using SMV, manpower, and net-good output.
7	Wash Planning	Wash/rework as value, quality, and capacity domain.
8	WIP and Queue Monitoring	Hidden waiting time and ownership made visible.
9	Exception and Alert Surface	Structured recovery action with owner and due date.
10	Shipment Readiness	Dispatch readiness separate from production completion.

7. MVP Scheduling Grain Decision

Decision: MVP scheduling grain should be order x style x color/shade-lot x production stage x workcenter/line x date/shift, with wash handled at wash-batch grain.

Operation-level detail should exist in master data and line-balance review, but the first scheduling engine should not attempt full operation-by-operation finite scheduling unless validated as necessary.

8. Scheduling Boundary Cases and System Reaction Rules

The system must define what happens when orders are cancelled, quantities change, capacity is lost or added, wash repeats, quality fails, shipment dates move, or frozen plans need to change. MVP should provide rule-based impact preview and recommended recovery actions, with planner approval for committed schedule changes.

Event	Stage	System Reaction	Planner Action	Audit / Approval
Order cancellation	Before procurement	Release planned capacity; cancel open plan intent.	Review commercial impact.	Audit.
Order cancellation	After cutting	Freeze or reclassify WIP; update load and shipment risk.	Management disposition decision.	Approval required.
Capacity loss	Sewing	Recalculate load, risk, and affected orders.	Reassign, split, or recover.	Audit.
Capacity loss	Wash	Recalculate wash queue, rewash reserve, and shipment risk.	Resequence, overtime, split batch.	Approval if committed.

Event	Stage	System Reaction	Planner Action	Audit / Approval
Wash rework	Post-wash QC	Create rewash WIP and load.	Approve and resequence.	Audit.
Shipment pull-in	Any stage	Recalculate priority, capacity conflict, and recovery options.	Prioritize, simulate, approve.	Approval if override.
Frozen plan change	Firm/frozen zone	Create plan-change request and impact preview.	Approve, reject, or revise.	Approval required.

Part A: MVP Surface Requirements

9. Order Lifecycle Surface

Provide one digital thread for every customer order from enquiry/quotation visibility through shipment.

Key capabilities

- Order master view with customer, buyer, style, PO, quantity, delivery date, current stage, risk, owner, and next action.
- Upstream milestone visibility: enquiry, quotation, sample request, sample submitted, sample approved, order confirmed.
- Order change status: cancellation, quantity change, delivery change, and split shipment request.
- Product complexity class, wash complexity class, and boundary-case flag.

Success criteria

- Every order has one system-level lifecycle status.
- Delayed or changed orders have visible reason, owner, and next action.

10. PCD Readiness Surface

Enforce production readiness before cutting starts.

Key capabilities

- Validate PO, BOM, fabric receipt, fabric QC, shade lots, shrinkage report, trims, pattern, marker, PP sample, wash standard, line allocation, wash capacity, and QC file.
- Add readiness checks for product complexity class, wash route, dry/wet requirement, fashion-effect standard, post-wash QC criteria, rewash handling rule, and capacity impact.
- Support Ready, Conditionally Ready, Blocked, and Escalated statuses.

Success criteria

- No order is released to cutting unless wash route and complexity class are visible.
- Conditional release includes reason, owner, expiry, and recovery action.

11. Weekly Planning Workbench

Convert the order book into an executable weekly cross-functional plan.

Key capabilities

- 2-4 week firm plan and 8-12 week rolling visibility.
- Order sequencing, cutting plan, sewing allocation, wash plan, finishing, inspection, shipment, and load vs capacity view.
- Frozen-zone and firm-zone plan controls.
- Impact preview for order change, cancellation, capacity loss, and shipment pull-in.

Success criteria

- Weekly plan becomes the coordination contract across departments.
- Plan cannot freeze while unresolved material, PCD, capacity, or wash conflicts remain.

12. Daily Production Release Surface

Release only what is actually ready and executable today.

Key capabilities

- Validate material, previous process, machine, manpower, quality, workcenter capacity, and next-process readiness.
- Check release impact against downstream wash capacity.
- Support release, block, exception approval, daily release history, and boundary-case holds.

Success criteria

- Daily release becomes gate-based.
- Release does not create downstream wash overload without explicit warning.

13. Workcenter Load Monitor

Show load, capacity, queue, WIP ageing, utilization, and active constraints across workcenters.

Key capabilities

- Define capacity unit, planning bucket, primary and secondary constraint resource, normal capacity, overtime capacity, capacity loss triggers, and recovery levers.
- Monitor Fabric QC, Cutting, Sewing, Dry process, Wet wash, Drying, Finishing, Packing, Final QC, and Shipment documentation.

Success criteria

- Bottleneck shifts are visible early.
- Recovery action is triggered before shipment risk becomes critical.

14. Sewing Line Loading Surface

Support realistic sewing line allocation and production target setting.

Key capabilities

- Assign orders to lines and calculate daily target from SMV, manpower, machines, skill, absenteeism, efficiency, learning curve, and quality performance.
- Distinguish gross target, quality-adjusted target, and net-good output.
- Handle absenteeism, machine breakdown, and style complexity changes as boundary cases.

Success criteria

- Sewing plan reflects realistic net-good capacity.
- Line loading decisions improve resource utilization.

15. Wash Planning Surface

Plan wash/laundry as a denim value-creation, quality-risk, and capacity-constraining domain.

Key capabilities

- Define wash complexity class, fashion-effect requirement, dry process, wet wash route, repeat-cycle possibility, and rewash reason.
- Create wash batches, maintain shade-lot identity, track batch queue, track completion, reserve repeat-cycle capacity, and link outcome to finishing release.
- Capture cycle count, post-wash shade/effect QC outcome, capacity impact, and wash route performance analytics.

Success criteria

- Rewash is visible as capacity-consuming work, not informal rework.
- Post-wash quality outcomes update WIP, capacity, and shipment risk.
- Wash queue overload triggers exception and recovery recommendation.

16. WIP and Queue Monitoring Surface

Monitor work waiting between processes.

Key capabilities

- Track WIP quantity, ageing, hold reason, next process, owner, shipment risk, threshold, and escalation trigger.
- Track shade-lot and batch identity where relevant.
- Distinguish normal queue from quality hold, rework hold, and capacity hold.

Success criteria

- Hidden queues become visible.
- Ageing WIP escalates into the Exception and Alert Surface.

17. Exception and Alert Surface

Drive exception-based planning and recovery action.

Key capabilities

- Capture exception type, affected order, severity, owner, due date, suggested action, status, and escalation level.
- Add alerts for cancellation after cutting, capacity loss, overtime capacity addition, repeat wash beyond threshold, post-wash rejection, shipment pull-in, frozen plan change, and partial shipment.
- Use rule-based recommendations for recovery actions.

Success criteria

- Critical issues are not buried in WhatsApp or Excel.
- Every critical issue has owner, due date, and visible action history.

18. Shipment Readiness Surface

Protect final shipment commitments.

Key capabilities

- Track shipment date, finished quantity, packed quantity, short quantity, final QC, AQL, cartons, labels, documentation, booking, and split shipment decision.
- Link shipment readiness to boundary-case events.
- Track premium freight or extra cost where applicable.

Success criteria

- Production completion and dispatch readiness are clearly separated.
- Shipment risk is visible before dispatch date.

Part B: Mature-State Scope

The MVP stabilizes planning discipline. Mature deployment extends into deeper commercial, technical, execution, governance, and analytics capability.

Surface	Aligned Mature-State Intent
Executive Control Tower	Management risk dashboard for shipment risk, bottlenecks, utilization, overtime exposure, recovery actions, and order-book health.
Enquiry and Costing	Full costing and feasibility surface remains mature-state; MVP only captures milestone visibility unless business confirms immediate need.
Sampling and Approval Tracker	Full sample workflow remains mature-state; MVP captures sample approval as lifecycle and PCD dependency.
Style Master and Technical File	Maintain wash complexity class, fashion-effect requirement, route family, repeat-cycle probability,

Surface	Aligned Mature-State Intent
	operation bulletin, and quality checkpoints.
BOM and Material Planning	Material requirement calculation, fabric width and shrinkage impact, shortage alerts, and procurement linkage.
Procurement and Vendor Follow-up	Vendor PO tracking, ETA, delay alerts, criticality ranking, and scorecard.
Fabric Inward and Fabric QC	Roll-wise receipt, shade lot, 4-point inspection, width, GSM, shrinkage, skewing, stretch, and pass/fail/hold.
Operator Skill and Capacity	Skill matrix, efficiency, quality rating, absenteeism, learning curve, and recommended allocation.
Cutting Room	Cutting plan, shade-wise sequence, marker approval, spreading, cut quantity, bundle numbering, and issue to sewing.
Wash Recipe and Batch Execution	Detailed recipe execution, machine assignment, parameters, shade/hand-feel/measurement result, and batch approval.
Quality Management	QC across all stages with defect category, root cause, and quality-adjusted capacity feedback.
Rework and Recovery	Rework order, affected quantity, responsible process, recovery time, capacity reservation, shipment impact, and closure.
Calendar / Gantt Planning	Visual order, department, line, wash, shipment, approval, and procurement timelines.
Recovery Planning / What-If Simulation	Scenario tests for fabric delay, line split, wash rework, overtime, shift extension, and shipment pull-forward.
Plan Change and Approval	Frozen-zone and firm-zone plan change request, reason code, impact analysis, approval, and audit trail.
Department Handover	Formal interdepartmental handovers with completeness checks.
Master Data Governance	Completeness and version control for style, BOM, SMV, workcenter capacity, calendar, wash recipe, and vendor lead time.
Performance Analytics	OTIF, utilization, plan adherence, line efficiency, net-good output, wash rework, quality defects, WIP ageing, and recovery cost.
Mobile / Shopfloor Update	Low-friction output, defect, shortage, machine issue, absenteeism, and handover updates.
Role-Based Home Surfaces	Role-specific dashboards for management, planners, merchandisers, procurement, QC, production, and shipment teams.

Part C: Functional Requirements

Req. ID	Requirement
FR-001	System shall maintain order lifecycle status from confirmation to shipment.
FR-002	System shall maintain planned and actual dates for each major order stage.
FR-003	System shall enforce PCD readiness checks before cutting release.
FR-004	System shall support weekly capacity-based production planning.
FR-005	System shall support daily production release based on readiness.

Req. ID	Requirement
FR-006	System shall monitor load and capacity at each workcenter.
FR-007	System shall identify current and future bottlenecks.
FR-008	System shall support sewing line loading using SMV, manpower, and capacity assumptions.
FR-009	System shall distinguish gross output from net-good output.
FR-010	System shall support wash route and wash batch planning.
FR-011	System shall track rewash and repeat wash capacity consumption.
FR-012	System shall track WIP quantity and ageing between processes.
FR-013	System shall generate exceptions for material, capacity, quality, approval, WIP, and shipment risks.
FR-014	System shall assign owner and due date to every exception.
FR-015	System shall track shipment readiness separately from production completion.
FR-016	System shall maintain audit trail for plan changes and conditional releases.
FR-017	System shall support role-based views and permissions.
FR-018	System shall support integration with ERP or existing order/material data sources.
FR-019	System shall support export and reporting for management review.
FR-020	System shall preserve historical data for performance analytics.
FR-021	System shall maintain product complexity class for denim and chino route planning.
FR-022	System shall define MVP scheduling grain across order, style, color/shade-lot, stage, workcenter/line, date/shift, and wash batch.
FR-023	System shall support rule-based impact preview for order cancellation, quantity change, shipment date change, and capacity change.
FR-024	System shall treat wash route, dry/wet step, post-wash QC, and rewash cycle as first-class planning events.
FR-025	System shall capture rewash reason, affected quantity, expected time, capacity impact, and approval status.
FR-026	System shall maintain business-operational capacity definitions by workcenter.
FR-027	System shall distinguish planner-controlled recommendations from automatic system changes.
FR-028	System shall capture upstream milestones for enquiry, quotation, sampling, technical file receipt, and order confirmation.
FR-029	System shall maintain audit trail for boundary-case events and scheduling reaction decisions.
FR-030	System shall support seed scenarios or simulations for cancellation, capacity loss, rewash, and shipment pull-in before full planning-engine rollout.

Part D: Business Rules

PCD Release Rules

- No order should move to cutting without PCD readiness status.

- Blocked orders must have open dependency and owner.
- Conditional release must require reason, owner, expiry, and approval.
- Fabric QC failure should block PCD unless overridden by authorized role.

Planning Rules

- Weekly plan must be capacity checked before freeze.
- Frozen-zone changes must require reason code and approval.
- Daily release must validate real readiness.
- Workcenter overload should trigger exception.
- Orders with high shipment risk should receive priority review.

Capacity Rules

- Capacity should be based on agreed unit, planning bucket, available minutes, manpower, machine availability, SMV, efficiency, and demonstrated output.
- Net-good output should account for quality loss and rework.
- Rework and rewash must consume capacity.
- Wash capacity must be separately planned from sewing capacity.

Scheduling Grain Rules

- MVP scheduling shall operate at order x style x color/shade-lot x stage x workcenter/line x date/shift grain.
- Wash shall be planned at wash-batch grain.
- Operation-level data shall support SMV, line balance, and technical validation, but full operation-by-operation finite scheduling is outside MVP unless later validated.

Boundary-Case Rules

- Order cancellation, quantity change, delivery-date change, capacity change, rewash, QC failure, and shipment pull-in must create a visible system event.
- Each event must show impacted orders, affected capacity, shipment risk, suggested recovery, owner, and approval requirement.
- MVP should recommend and preview impact; it should not automatically reschedule frozen plans without approval.

Rewash / Repeat Cycle Rules

- Rewash must consume planned capacity.
- Rewash must carry reason, affected quantity, approval owner, expected duration, and shipment impact.
- Repeated wash beyond threshold must trigger exception.
- Post-wash QC result must control release to finishing.

Automation Boundary Rules

- MVP shall use rule-based impact preview and recommendations.
- Planner or authorized approver shall control plan changes in frozen or firm zones.
- Fully autonomous rescheduling is outside MVP unless validated by the business.

Shipment Rules

- Shipment readiness must include final QC, inspection, packing, documents, and booking.
- Production complete does not mean shipment ready.
- Short shipment risk must be visible before shipment date.
- Split shipment decision must be tracked and approved.

Order Cancellation Rules

Cancellation Stage	Required System Reaction
Before procurement	Release planned capacity, cancel open procurement intent, and audit decision.
After procurement	Flag material liability, review reallocation, and update capacity and cost exposure.
After cutting	Freeze or reclassify cut WIP and require management decision.
During sewing	Update WIP, capacity, and shipment plan; require planner action.
During/after wash	Flag finished/semi-finished goods disposition and customer impact.

Part E: Non-Functional Requirements

Category	Requirement
Usability	Screens must be simple, action-oriented, and suitable for planners and shopfloor supervisors.
Performance	Workcenter load, WIP, and order-risk views should load quickly for large order volumes.
Scalability	System must support large factory operations with many lines, workcenters, styles, and orders.
Reliability	Planning data must be version-controlled and auditable.
Security	Access should be role-based and protect sensitive commercial and customer data.
Auditability	Plan changes, overrides, conditional releases, and shipment changes must be traceable.
Configurability	Workcenters, statuses, thresholds, calendars, roles, product complexity, and capacity units must be configurable.
Explainability	Planning recommendations must show why the system is recommending a recovery action.
Data Freshness	Planning-critical screens must show data source and last update status.
Scenario Traceability	Boundary-case outcomes must be traceable to the rule or user decision that caused them.
Change Governance	Frozen-plan changes, overrides, and conditional releases must require controlled approval and

Category	Requirement
	audit.

Part F: Integration Requirements

Integration Area	Purpose
Order Management / ERP	Order, customer, style, PO, quantity, delivery date.
BOM / Material System	Fabric, trims, labels, packing materials, and consumables.
Procurement	Vendor PO, ETA, inward status, delay alerts.
Sampling / Approval Records	Upstream milestone visibility and PCD readiness dependency.
Fabric QC	Inspection status, shade lot, shrinkage, and pass/fail/hold results.
Production Actuals	Cutting, sewing, wash, finishing output.
Quality System	Defects, holds, rework, AQL, post-wash QC.
HR / Attendance	Operator availability and absenteeism.
Machine / Maintenance	Machine availability, downtime, and breakdown.
Shipment / Logistics	Packing, inspection, dispatch, documents, booking.
FastReact / Existing Planning Tool	Define replacement, coexistence, import, or export role clearly.
Wash Route / Recipe Master	Route family, standard cycle, repeat-cycle rules, and quality checkpoints.
Order Change / Cancellation Source	Capture cancellation, quantity change, and delivery-date change events.
Capacity Calendar	Shift, overtime, downtime, absenteeism, and approved capacity changes.

Part G: Roles and Users

Role	Key Usage
Management	Control tower, shipment risk, bottlenecks, performance, approval escalation.
Production Planner	Weekly plan, daily release, workcenter load, boundary-case recovery.
Merchandiser	Order lifecycle, approvals, buyer comments, upstream milestones.
Procurement User	Material status, vendor follow-up, shortage alerts.
Fabric QC User	Fabric inspection, shade lot, shrinkage, and clearance.
Cutting Manager	Cutting readiness, cut output, bundle handover.
Sewing Manager	Line load, output, WIP, constraints, underperformance.

Role	Key Usage
Washing Manager	Wash plan, batch status, rewash, post-wash QC linkage.
Finishing Manager	Finishing queue and packing readiness.
QC Manager	Quality holds, inspection, defects, post-wash/final QC.
Shipment Team	Shipment readiness, documents, booking, dispatch, split shipment.
Shopfloor Supervisor	Daily updates, output, issues, handover.

Part H: Reporting Requirements

Report	Purpose
Order Status Report	End-to-end order progress.
PCD Readiness Report	Orders ready, blocked, conditional, or escalated.
Weekly Plan Report	Department-wise weekly plan and freeze status.
Daily Release Report	Work released, blocked, and conditionally released.
Workcenter Load Report	Load vs capacity by workcenter.
Sewing Line Load Report	Line-wise plan, target, output, and net-good result.
Wash Queue Report	Wash load, queue, rewash, route, and post-wash QC.
WIP Ageing Report	Stuck WIP, hold reason, owner, and shipment risk.
Exception Report	Open issues and recovery actions.
Shipment Readiness Report	Orders ready or at risk for shipment.
Scheduling Boundary Case Report	Cancellation, capacity change, rewash, and shipment-pull events with system reaction.
Wash/Rewash Performance Report	Wash queue, route performance, rewash count, reason, and capacity impact.
Capacity Change Log	Approved and unapproved capacity changes by workcenter.
Order Change Impact Report	Impact of cancellation, quantity change, and delivery-date change.
Planning Rulebook Compliance Report	Whether critical releases followed defined scheduling rules.

Part I: Implementation Roadmap

Phase 0A: Scheduling Behaviour Rulebook and Simulation Validation

Before deep implementation, conduct a focused business workshop and produce the following deliverables:

14. Capacity definition matrix.
15. MVP planning grain decision.
16. Order cancellation matrix.
17. Capacity change matrix.
18. Wash/rewash behaviour matrix.
19. Shipment risk reaction matrix.
20. Planner-control vs system-control boundary.
21. Boundary-case seed scenario pack.
22. BRD-to-build traceability matrix.

Build gate: Do not finalize daily release, workcenter load, or wash planning logic until Phase 0A confirms capacity definitions, wash/rework behaviour, and boundary-case reaction rules.

Phase 1: MVP Foundation

- Order Lifecycle
- PCD Readiness
- Weekly Planning Workbench
- Daily Production Release
- Workcenter Load Monitor
- Sewing Line Loading
- Wash Planning
- WIP and Queue Monitoring
- Exception and Alert Management
- Shipment Readiness

Phase 2: Material, Sampling, and Technical Master Expansion

- Enquiry and costing
- Sampling and approvals
- Style technical file
- BOM and material planning
- Procurement and vendor follow-up
- Fabric inward and QC

Phase 3: Shopfloor and Quality Depth

- Cutting room
- Wash recipe execution
- Quality management
- Rework and recovery
- Operator skill and capacity

- Department handover
- Mobile shopfloor updates

Phase 4: Governance, Simulation, and Analytics

- Executive control tower
- Calendar / Gantt planning
- What-if simulation
- Plan change approval
- Master data governance
- Performance analytics
- Role-based home surfaces

Part J: Risks and Mitigation

Risk	Mitigation
Users continue using Excel	Make MVP surfaces action-critical and reduce duplicate entry.
Poor master data	Start with minimum required master data and completeness checks.
Delayed shopfloor actuals	Use simple update flows and supervisor-level capture.
Resistance to gate-based release	Use management-backed policy for PCD and daily release.
FastReact overlap confusion	Define integration or coexistence role clearly.
Wash complexity under-modeled	Reframe wash as value-creation, quality-risk, and capacity domain; include in MVP.
Scheduling grain remains ambiguous	Add MVP scheduling grain decision before build.
Boundary cases are discovered too late	Add Scheduling Behaviour Rulebook and boundary-case test pack.
Premature automation creates user distrust	Use rule-based recommendation and planner approval in MVP.
Pre-order scope expands MVP too far	Capture upstream milestones in MVP; defer full costing unless confirmed.
Chinos over-modeled as denim	Use product complexity classes and route configuration.
Too many alerts	Prioritize shipment-impacting exceptions.

Part K: Open Decisions and Workshop Questions

Scope and Trigger

23. Was the original business trigger quotation/costing, production scheduling, delivery reliability, FastReact underuse, or all of these?
24. Is the immediate pain before order confirmation, after order confirmation, or both?
25. Are chinos handled in the same flow or a simpler production route?

Current Planning

26. What planning is done in FastReact today?
27. What planning is still done in Excel?
28. Who owns the weekly plan and daily release?
29. What is the current planning grain: order, PO, style/color, batch, line, operation, or shipment?

Wash and Denim

30. What are the main wash route families?
31. How is dry process planned separately from wet wash?
32. How often does rewash occur?
33. What are the main rewash reasons?
34. Who approves rewash?
35. Is drying or post-wash QC a separate bottleneck?

Capacity

36. How is sewing capacity measured?
37. How is wash capacity measured?
38. How is absenteeism reflected?
39. How is overtime approved?
40. How are machine breakdowns captured?
41. Which recovery levers are acceptable by workcenter?

Boundary Cases

42. What happens when an order is cancelled before procurement, after fabric receipt, after cutting, or during wash?
43. Should the system automatically reschedule or recommend actions?
44. Who can approve frozen plan changes?

Prototype Confidence

45. What minimum prototype will give the business confidence?
46. Which real orders should be used as seed simulations?
47. Which process should be simulated first: sewing, wash, or shipment risk?
48. Which boundary cases must be proven before committed build?

20. Final BRD Summary

Eratex requires an end-to-end planning and scheduling tool because its current planning environment appears to be partially digitized but still operationally dependent on Excel. The aligned BRD sharpens the root cause: denim manufacturing requires a governed operating spine that understands wash/laundry complexity, WIP movement, capacity changes, boundary cases, and exception ownership.

The proposed tool must become the daily operating discipline layer for Eratex. It must enforce order visibility, PCD readiness, weekly and daily planning control, workcenter load monitoring, realistic sewing line loading, denim-specific wash and rewash planning, WIP ageing visibility, exception ownership, shipment readiness, and controlled system reaction when factory reality changes.

Final position: Build a governed denim-aware operating spine, not just a scheduling screen.